

Flexible Work Arrangements and Firm Outcomes: Experimental Evidence from India

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Preliminary and incomplete.

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Abstract

This paper studies the impact of flexible work arrangements on worker and firm output in the context of a large apparel manufacturing cluster in India. Production is organized as batch processing lines, where tasks are performed sequentially to produce the finished product. In the status quo work arrangement, workers choose their own work timings, resulting in significant dispersion in arrival times at the firm. I implement an experiment with 97 small factories and 600 workers, where all workers in treatment firms are incentivized to arrive at work by the owners' preferred start time for a period of 4-weeks. Workers are paid a daily bonus equivalent to 15% of average earnings if they arrive on or before the start time. Incentives increase the likelihood that workers arrive on or before the preferred start time in treatment firms by 40% relative to control firms. This results in large improvements in coordination at the firm level – dispersion in worker arrival times decreases by 35%. There is no effect on worker attendance and a modest increase in daily work hours of 30 minutes. Weekly firm output increases by 29%, resulting in a 12% increase in firm revenues. These effects can be explained by two mechanisms: first, incentives reduce uncertainty in labor supply to the firm, encouraging owner's to expand the firm's capacity. Treatment firms receive 16% more new materials to produce, suggesting that the increase in output was not only driven by existing inventory. Second, I show suggestive evidence of positive spillovers along the production line within firms.

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1 Introduction

As countries develop, they undergo structural changes in the organization of labor – in particular, a shift from self-employment to wage-jobs in firms (Bandiera et al., 2022). However, the take-up of wage jobs in many low-income contexts still remains low. Moreover, among those who do take-up wage-jobs, rates of separation and absenteeism remain high, suggesting that the type of wage-jobs offered in such contexts may not match workers’ job preferences (Breza and Kaur, 2025). Understanding frictions to wage-employment is thus a crucial topic of research in developing countries. Recent work attributes these low rates of wage-employment to workers’ preferences for flexibility (Cefala et al., 2024; Oh et al., 2024).¹ While flexibility is an important job attribute for workers – providing flexibility may be costly for firms, especially in settings where the production process requires a high degree of coordination among workers (assembly line manufacturing). This creates a tension between worker and firm preferences, and may have broader implications for firm growth and economic development if coordination costs increase in scale (Lucas, 1978).

In this paper, I investigate this trade-off between firm and worker preferences for flexible work arrangements, focusing on flexibility in arrival times at work as a dimension of flexibility. I study the context of a large apparel manufacturing cluster in Tirupur, India which features many small-medium firms employing 5-15 workers (tailors). Firms are organized as batch processing lines, where each worker (tailor) performs a distinct operation on the garment in batches of 80-100 pieces. The status quo work arrangement in this setting reflects the trade-off between firm and worker preferences. Tailors are hired on piece rate contracts and organize as independent contractors in firms. This enables them to choose their work timings, instead of being required to follow uniform work shifts like their counterparts working in large/export firms.² Effectively, this means that workers do not coordinate their work schedules – in data from 97 firms and 600 workers, the difference in arrival times from firm’s stated start time ranges from 3 hours early to 5 hours late, with a 90-10 gap of almost 2 hours (see Figure 1). Moreover, because workers are only incentivized on individual output as opposed to team output (e.g. only 100 sleeves + 100 collars versus 100 pieces of the completed garment), workers’ individual choice of work schedules may create a coordination externality on both other workers and consequently the firm owner.

On the other hand, the production technology implies that worker coordination may improve production. *Because* operations need to be performed sequentially to produce the final output,

¹For example, Cefala et al. (2024) document flexible work schedules as the second most important reason (after earnings per day) for choosing a casual job over a regular job.

²In a separate survey of tailors and firm owners in Tirupur, I document large and systematic differences in work arrangements by firm size – see Table A4. Interviews with both small and large firm owners suggests that differences in the provision of flexibility arise from a combination of institutional and technological constraints: (a) labor laws and social compliance from international buyers require large/export firms to systematically record work shifts in order to provide workers government benefits and overtime pay, and (b) large/export firms are more likely to organize production in assembly lines, while small firms follow batch production processes. Although both forms of production involve a high degree of task specialization among workers (i.e, workers perform only 1 distinct task per product/day), assembly lines have a stronger requirement for worker coordination than batch production processes.

the lack of coordination in worker’s arrival times can either (a) limit worker’s ability to specialize on their expert task because they need to fill in for a late worker who usually performs the preceding task, or (b) increase idle time if they need to wait for the preceding task to be completed. Thus, such flexible work arrangements may slow the overall speed of production, and lower productivity by limiting specialization. This is consistent with owner’s beliefs about how flexibility affects production, and why imposing fixed work schedules may increase production: at baseline, owners were asked what type of measures related to their workers would help increase production (un-prompted). Nearly 90% of owners reported that they would like their workers to start work at the same time. This far outweighs other measures – 38% of owners would like workers to take fewer/shorter breaks, 25% would like workers to take fewer un-announced leaves and 17% to not bring their children to work (Figure 2a). When asked why coordinated timings would help increase production, 66% of owners suggest a reduction in idle time, and 51% suggest increased task specialization as possible mechanisms (Figure 2b). Additionally, 40% owners also allude to uncertainty about whether and when workers will arrive at the firm as another mechanisms. Coordinated timings can help owners plan and deliver orders on time, which helps them receive more orders in the future.

I conduct an experiment with 97 firms and ≈ 600 workers to test whether flexibility in work schedules affects firm productivity. I incentivize workers in treatment firms to follow fixed work start times for a period of 4-weeks. Treatment incentives are designed as a daily bonus of Rs. 100 (15% of average daily earnings), paid conditional on arriving at the firm by the owners’ pre-specified work start time (hereafter, preferred start time), elicited at the time of baseline. Thus, the incentive encourages both attendance and before-time arrival. Workers in control firms receive unconditional incentives during the treatment incentive period, matched to a randomly chosen treatment firm each week to address concerns about income effects on worker output.

I collect daily data on worker attendance, arrival times, breaks taken, and production starting 2-weeks before the incentives, and the 4-week incentive period. I define a worker as compliant with the treatment if they are present *and* arrived on or before the preferred start time. First, I estimate the effect of the incentives on worker attendance and compliance. I interpret the treatment effect on worker compliance as an upper-bound estimate of workers’ willingness to pay for workplace flexibility. Second, I study the effect of improved compliance on firm and worker output and (other economic measure). Finally, I empirically test two hypothesized channels of increased firm three: (1) an improvement in the allocation of tasks to workers – either through a decrease in multitasking, or better alignment of tasks to workers’ expertise, (2) spillovers along the production line to downstream workers, and (3) an increase in firms’ capacity.³

I find that treatment significantly increases the probability that workers arrive at work at or before owners’ preferred start times as elicited at baseline. This effect is large – the fraction of worker days that workers are present and arrive before the preferred time increases by 18 percentage

³This version of the draft provides suggestive evidence for a subset of these mechanisms.

points, a 43% increase relative to a control mean of 41% before the incentives were active. I find no effect on worker attendance, suggesting that the effect on worker compliance is only driven by changes in workers' arrival times. The effect on compliance is almost entirely driven by workers who consistently arrived after the preferred start time at baseline (hereafter, "late" workers), and female workers, who are more likely than male workers to be late at baseline. Workers in treatment firms on work on average 30-45 minutes longer than control firm workers, but this effect is small in magnitude and not statistically significant for the average worker. Similarly, there is no change in the number and total duration of breaks to compensate for arriving early. Overall, these changes in worker compliance reduce dispersion in worker arrival times at the firm level - total dispersion in worker arrival times, measured as the difference in arrival times between the earliest and the last worker, decreases by 15 minutes (11%). The deviation from preferred start time for the 90th percentile worker - which indicates the 2nd to last worker to arrive given firms have an average of 6 tailors - decreases by 31% relative to the control mean. Thus, incentives are successful in improving coordination in worker arrival times in treatment firms.

Next, I estimate the effect of incentives on overall firm output, measured as the total number of finished pieces produced by all workers in the firm. I find a statistically significant, 0.24 standard deviation increase in weekly firm output - 29% increase relative to the control mean of 6,900 and control standard deviation of 7,200, equivalent to an increase in total weekly output by $\approx 1,500$ pieces. Weekly firm revenues increase by INR 12,000 (12% relative to the control mean), while total labor costs, *including* the incentive paid to treatment workers, increases by INR 6,200 (15% relative to the control mean). Assuming that other non-labor costs did not change differentially between control and treatment firms, these results provide suggestive evidence that improving worker coordination increases firm output and profits.

To unpack what drives the increase in total firm output, I estimate the intent-to-treat effects of the incentive on *worker* output and earnings. Worker output is the sum of total pieces produced by a worker, across all tasks/products, regardless of whether the garment has been fully stitched. Total worker earnings are calculated as the sum of earnings for each task performed by the worker, and summed across all tasks. I find that weekly worker output increases by 0.35 standard deviations (35% relative to the control mean, equivalent to 1,100 more pieces per week). Weekly worker earnings (*excluding* the incentive amount) increase by 0.27 standard deviations (24% relative to the control mean). The magnitude of the earnings increase is equivalent to INR 700 - slightly larger than the maximum weekly incentive amount (INR 600) offered to workers in treatment firms.

Surprisingly, unlike the results on labor supply, increases in weekly worker output and earnings are realized by all workers, and not driven by female or late workers. I explore two channels that can explain these results: first, by changing worker arrival times (and consequently, both the order in which workers arrive *and* better coordination in worker arrival times), treatment may have resulted in a change in task allocation between workers in the firm. This may result in an increase in the extent of task specialization, or a better assignment of tasks to workers' expertise. Second, the fact

that workers for whom incentives were not binding also benefit from incentives suggests the presence of coordination externalities along the production line. For example, increased compliance among late and/or female workers increases their own output each day, but may have spillover benefits to downstream workers (workers who perform the task next in the order). The firm may realize increases in production through the production line even without changes in the task allocation, or changes in the degree of task specialization.

Finally, I find that treatment firms receive 0.19 SD larger quantity of new orders during the incentive weeks, equivalent to a 16% increase in new orders received relative to control firms. This is consistent with firm owners' reports at endline that they actively tried to seek new, more profitable orders from both existing and new buyers. Anecdotally, treatment firms owners are 26 percentage points (96%) more likely to agree that total production at their firm increased during the incentive weeks, and are twice as likely to attribute this increase to workers arriving on time.

This paper makes three contributions to the literature. The findings of this paper highlight an important, yet under-explored relationship between the provision of flexible work arrangements, and labor supply preferences (Mas and Pallais, 2020). I experimentally estimate workers' demand for flexible work arrangements and highlight the trade-off between worker preferences for flexibility and firm output. Workers' demand for flexibility has been documented across several different contexts (Mas and Pallais, 2017; Ho et al., 2024; Baysan et al., 2024; Wiswall and Zafar, 2018). This paper's main contribution is to link worker's WTP for flexibility with firm outcomes. I show that while workers individually prefer flexible work arrangements (as demonstrated by (a) the status quo dispersion in arrival times and (b) heterogeneity in response to the incentives), on average, all workers in the firm experience an increase in output and earnings when they arrive by the preferred start time. These findings are consistent with the idea that despite fixed-schedules being a dis-amenity for workers individually, there are potential gains to themselves *and* the firm.

This paper also speaks to a large literature on barriers to firm growth in developing countries. I build on previous work documenting search and hiring frictions in low-income countries (De Mel et al., 2019; Hardy and McCasland, 2023; Bassi and Nansamba, 2022; Caria and Falco, 2024; Carranza et al., 2020) by documenting that even conditional on forming matches, firms may face barriers to coordinating workers. This could have wider economic consequences on firms' ability to scale their workforce as coordination becomes costlier with scale (Lucas, 1978). My results suggest that improving coordination in worker arrival times at the firm do increase firm output and revenues, suggesting that frictions to coordinating workers can pose a significant barrier to firm growth in developing countries.

I contribute to the growing literature on the internal organization of small, informal firms in developing countries, building on recent work such as Bassi et al. (2022, 2023). I document worker preferences as the source of coordination costs for SMEs in developing countries, and how they affect firms' ability to specialize.

The remainder of the paper is organized as follows: [section 2](#) describes the study context and sample firms, [section 3](#) describes the experimental design, implementation and data collection, and [section 4](#) lays out the empirical strategy. [section 5](#) presents the main results and [section 6](#) concludes.

2 Study Context and Sample

The context of my study is Tiruppur district in the state of Tamil Nadu, India. This region is part of a major garment manufacturing hub in India, catering to both export and domestic markets since the 1970s. The production process in Tiruppur is highly decentralized and carried out by sub-contracted units, called job-work units – these units will form my study sample. Job work units are *small* factories, employing about 10-30 workers, at least 50% of whom are women.⁴

Firms are organized as batch processing lines, where each worker performs a distinct operation on the garment in batches of 80-100 pieces. Operations need to be performed sequentially in order to produce a finish piece of garment. Firms typically specialize in broad product categories - such as t-shirts, bottomwear, innerwear, or kids wear. The average weekly order size in these firms is about 4,000 pieces, varying with product type.

The study takes place in 4 neighborhoods located in Tiruppur North sub-district, Tiruppur, Tamil Nadu. These neighborhoods were identified as high density clusters of small-medium job-work units during prior scoping work and area mapping exercises. I used the following criteria to screen firms to be part of the sample:

1. Has a stitching section
2. Employs at least 2 tailors who are not related to the firm owner or who work for pay (piece rate contract)
3. Must deliver finished products back to the buyer/retailer
4. Must receive cut-pieces of cloth to be stitched (ie does not only produce own designs)

All firms that satisfy these criteria were eligible for baseline. In case eligible firms were located adjacent to each other, I approached alternate firms in order to minimize spillovers between treatment and control firms located next to each other.

3 Experiment Design

Firms are randomly assigned to one of two treatment groups - (T) Treatment ($n = 50$), and (C) Control ($n = 47$). All workers in treatment firms were eligible to receive a daily bonus of Rs.

⁴Note that these firms are of course larger than the typical small developing country firm; however they employ a substantial fraction of the workforce within the textile industry in Tiruppur district - as of 2013, the share of total employment in textiles in 10-30 worker firms ranged between 20-30% in rural and urban areas in Tiruppur district (source: Economic Census, 2013-14). Such organization structures are common in many industrial clusters around the world – see [Nadvi and Schmitz \(1994\)](#); [Banerjee and Munshi \(2004\)](#); [Chaudhry and Woodruff \(2013\)](#); [Atkin et al. \(2017\)](#).

100, equivalent to 15% of their *average* daily earnings if they arrived at the firm and started work by the company’s preferred start time. Thus, the incentive encourages both on-time arrival and attendance. The total incentive amount was paid to workers as part of their weekly earnings.⁵ The unconditional incentive amount for control group firms was matched to the total realized payment made to a randomly selected treatment firm, capped at Rs. 300 per week (the maximum incentive amount in the low incentive treatment group T2). These treatment-control firm (randomized) pairings were created separately each week. [Figure 4](#) shows the experiment design.

3.1 Data and Summary Statistics

Baseline: I conduct a baseline survey with owners to capture owner and firm characteristics, and a complete worker roster, which is followed by a baseline survey of all workers at the firm. Randomization and treatment assignment was done after owner and worker baseline surveys.

Baseline firm and worker characteristics, and balance by treatment arms are reported in [Table 1](#) and [Table 2](#) respectively. On average, firms hire 6 tailors (excluding owners/co-owners), and have been operating for 6-7 years. Firms produce 4,000 pieces of output each week, and earned a profit of INR 30,000 in the last 30 days (USD 340).

Owners directly supervise workers (that is, without a manager/contrator) in almost all firms – owners also perform stitching work in 80% of firms, and work as regular tailors in about 60% of firms. About 80% owners are male, 56% have completed secondary school and have 17-19 years of experience in the garment industry.

Workers are on average 37 years old (similar to owners), 44% are male and most workers are married (94%) and have at least one child younger than 14 years of age (85%). 87% workers live within a 15 minute commute to the firm, have worked at the current firm for 2.5 years and have 15 years of experience in the garment industry. Worker characteristics are balanced across treatment and control firms ([Table 2](#)).

Order, Task and Piece Rates Data: At baseline, I also record a roster of all styles that the firm is currently producing. For each style, I record the piece rate that the owner receives for a finished stitched piece and its task structure – a list of all tasks which they need to be performed to produce the finished product, the order in which they are performed, and their piece rates. This roster is updated throughout the 6-week study period to incorporate new styles that the firm is producing.

Daily Production and Attendance Data: Starting 1-2 weeks after baseline, I collect data *daily* for a period of 6-weeks, out of which workers are paid incentives for 4-weeks.⁶ Each day, I

⁵All workers who were working full-time at the firm at any point during the study were eligible to receive incentives. Workers’ full-time status was determined by the owner and verified by the research team using daily attendance data.

⁶In Phase 1, I collected data 1 week pre-incentives and 1 week post-incentives. In Phase 2, I collected 2 weeks of pre-incentives data only, dropping the post-incentive data collection. In Phase 3, I collected 2 weeks of pre-incentives

record worker’s attendance, arrival time, break times during the day, and daily output (by product-task). Arrival times and attendance are recorded by surveyor observation and are not prone to mis-reporting in order to game the incentive; remaining data is self-reported by workers. Daily output is recorded for all production workers, which includes the owner and/or co-owner in some firms. I use daily output data from all production workers to construct an independent measure of total firm production. In addition to tailoring/production output, I also record whether the worker did any other work at the firm, such as helper work, receiving or delivering raw material/finished pieces, machine repair, purchasing inputs, etc.

Table A1 summarizes key labor supply and production outcomes during the pre-incentive weeks, and differences across two worker types – gender, and whether the worker consistently arrived after the preferred start time during the pre-incentives weeks (hereafter, referred to as “late” versus “early” workers). Column (1) reports summary statistics for the overall worker sample; columns (2)-(3) report summary statistics for female and male workers, and columns (4)-(6) report summary statistics for early, “late”, and “very late” workers. “Late” and “very late” are indicators for workers whose average pre-incentive arrival times were between 0 and 30 minutes, and above 30 minutes of the preferred start time, respectively.

Female and male workers have similar demographics – the only exceptions being that women are more likely to be married, live closer to the firm, and have 5 years less experience in the industry than men. On the other hand, late and very late workers are more likely to be female, married, and have more children below 14 years of age. There are no differences in commute times between early, late and very late workers, suggesting that arrival times observed in status quo are not driven by commuting differences, but are likely driven by time commitments in the household before the start of the work day.

Pre-incentive attendance rates are similar across men and women, however, there are several notable gender differences in terms of arrival times and compliance: women arrived 21 minutes after the preferred start time on average, while men arrive 11 minutes before the start time – consequently, women were more likely than men to be “late” and “very late”, and 23 percentage points less likely to comply with the preferred start-time - one-third of men’s average compliance rate. On the other hand, attendance rates are the highest among late workers at 83%, 10 percentage points higher than attendance rates among early and very late workers.⁷ More surprisingly, only 65% of early workers, 45% of late workers, and 10% of very late workers complied with owners’ preferred start times before the incentives – suggesting that (a) early workers are not the most reliable workers, and (b) late workers do comply almost half of the time. Given these patterns, I

data, 4 week of incentives data and 1 week of post-incentives data.

⁷This implies an overall average absenteeism rate of 24%, and 17% among late workers. This range is similar to absenteeism rates documented in other studies. Notably, absenteeism rates among late workers are similar to those reported in large, export factories – 11% in Tanzania (Ho and Huang, 2024), and 15% in India (Adhvaryu et al., 2022). The average absenteeism is similar to those observed in casual labor markets, which are closer to the labor market in this study (25% in rural casual labor markets in Burundi (Cefala et al., 2024), and 20-30% in urban casual labor markets in India (Cefala et al., 2024).

expect the incentive to encourage higher attendance rates among early workers, and to encourage both attendance and on-time arrival among late and very late workers. These patterns also suggest a possible trade-off between regular attendance and on-time arrival. Finally, I observe that women seem to compensate for their late arrival by taking fewer number of breaks during the day, so that average daily hours are only 40 minutes (0.6 hours) lower than men. Late workers work 40 minutes more than both early and very late workers, with no striking differences in the number or duration of breaks taken.

In terms of production outcomes, workers perform 1-2 tasks per day on average, and work on non-tailoring tasks 12% of the time. The average piece rate of tasks performed in a day is 80 paise, with some notable differences by gender – women work on tasks with 30 paise lower average piece rates than men; and baseline lateness – average piece rates decline monotonically with baseline lateness. All workers produce 100 pieces per hours, resulting in small daily earnings differences – women earn Rs. 150 less per day than men, and very late workers earn Rs. 200 less per day than early workers. These differences in piece rates and earnings partly stem from the type of tasks performed – men, and early workers are most likely to perform Complex Singer tasks before the incentives. These tasks are slower and have higher piece rates than the average task.⁸

Deliveries Data: At the end of each week, I record all orders received and delivered by the firm in the past week. I use information on total quantity delivered in a week as a separate measure of firm output and sales in that week.⁹ In addition to information on deliveries, owners are also asked to match each delivery to the date that material was received – this allows me to identify distinct “orders” received by the firm. Additionally, for all new materials received, I elicit owners’ estimated planned delivery date, which I can compare with the actual delivery dates reported by owners to assess how treatment affects owners’ planning of production.

Endline: Finally, endline surveys conducted during the last week of the study capture firm owner’s and worker’s perceptions of the treatment (in treatment firms), preferences for flexible work arrangements, changes in firm operations during the study and time-use during and outside of work (relative to the pre-study period). These questions allow me to examine (1) the nature of workers’ commitments/restraints that drive their preference for flexibility, (2) how compliance with the incentive impacted these commitments/restraints, and (3) whether compliance shifted workers’ preferences for flexible (versus strict) work arrangements.

⁸About 50% of all singer tasks are complex tasks, and output per hour on Complex Singer tasks is half of that on other tasks.

⁹As discussed later in the results section, I use firm output constructed from production data is my preferred measure of firm output, since the quantity of deliveries may be affected by stock of finished pieces at the firm which I do not observe at the start of 6-weeks of daily data collection.

3.2 Implementation

The experiment was implemented in three phases - Phase 1 was implemented with 17 firms (9 treatment, 8 control) from November 2024 to January 2025. Phase 2 was implemented with 43 firms (22 treatment, 21 control) from March 2025 to May 2025. Phase 3 was implemented with 37 firms (19 treatment, 18 control) from August 2025 to October 2025. [Figure 5](#) shows the study timeline. Phase 1 only included high treatment firms (T1), and the half of the control firms did not receive any incentive (pure-control). In Phase 2, all control firms receive unconditional incentives matched to a randomly chosen treatment firm, up to the maximum allowable amount in low-treatment firms (Rs. 300). These changes were made to simplify field logistics by minimizing both distance between sample firms as well as firm attrition. In Phase 3, I increased the daily incentive amount in treatment firms to INR 150 per day.

Randomization for Phase 1 was conducted by generating a random order of treatment and control assignments, which firms were matched to in the order in which their baseline survey was conducted. Randomization for Phases 2 and 3 was stratified on the degree of dispersion in worker arrival times around the owner’s preferred work start time, using worker’s usual arrival times as reported by the owner at baseline. All specifications reported in the paper control for phase by strata fixed effects to account for the method of randomization in each phase ([Bruhn and McKenzie, 2009](#)).

4 Empirical Strategy

I estimate the effect of incentives on worker level outcomes using the following specification:

$$Y_{i,f,t} = \alpha + \beta T_{i,f} \times \mathbb{I}\{\text{During}\}_t + \lambda_{i,f} + \mu_w + \mu_{dow} + \varepsilon_{i,f,t} \quad (1)$$

where $Y_{i,f,t}$ is the outcome of interest for worker i in firm f at time t , and time is either day or week, depending on the outcome variable. $T_{i,f}$ is an indicator that equals 1 if a firm f is assigned to treatment (fixed over time), $\mathbb{I}\{\text{During}\}_t$ is an indicator variable that takes the value 1 if the study week is an incentive week, and $\lambda_{i,f}$ denotes worker fixed effects. μ_w are study-week fixed effects,¹⁰ and μ_{dow} are day-of-week fixed effects (included only in day-level specifications). β is the coefficient of interest, and standard errors $\varepsilon_{i,f,t}$ are clustered at the firm level. In the current analysis, I pool both high and low treatment arms.

I use a similar specification for firm-level outcomes:

¹⁰Study weeks take the value 1 to 4 for the 4 incentive weeks, 0 for the week prior to incentives. Weeks 0-4 are common across Phase 1 and 2. The first pre-incentive week in Phase 2 is coded week -1, and the final, post-incentive week in Phase 1 is coded week 5. In the current analysis, I drop the post-incentives week (week 5).

$$Y_{f,t} = \alpha + \beta T_f \times \mathbb{I}\{\text{During}\}_t + \lambda_f + \mu_w + \mu_{dow} + \varepsilon_{f,t} \quad (2)$$

where indices and variables are defined as above, and standard errors $\varepsilon_{f,t}$ are clustered at the firm level. Some firm outcomes are only measured at endline, in which case I will run [Equation 2](#) without the t subscript and without firm and study-week fixed effects. Day-of-week fixed effects are only included in firm-day level regressions.

5 Results

5.1 Labor Supply

First, I test the effect of the incentives on worker attendance and compliance with work start times. Columns (1) and (2) in [Table 3](#) report results for worker attendance and compliance with the treatment incentives. Compliance is defined as an indicator for being present at the firm, *and* arriving by the owner’s preferred start time. I find no effect on attendance (column (1)), and an 18 percentage point increase in compliance (column (2)) in treatment firms during the incentives weeks. This represents a 43% increase in compliance relative to the pre-incentive control mean (41%), and is driven by workers arriving on time, rather than an a change in worker attendance. The effect on compliance is larger, and more meaningful when conditioned on workers being present at the firm – on average, 57% workers - or 3-4 out of 6 workers - arrived on time conditional on being present at the firm. Treatment incentives increase worker compliance to 78% – this amounts to 4-5 out of 6 workers who arrive before the preferred time (conditional on being present). Given the small size of the workforce in these firms, the treatment encourages almost all workers to arrive before the preferred start time. This effect is visualized in [Figure 7](#), which shows that the effect of the incentives on worker arrival times was concentrated in the first 45 minutes of the start time.

Next, I examine whether these effects vary by gender, and baseline lateness in [Table 4](#). Treatment effects on attendance for women and “late” workers are no different than their respective base categories, consistent with the result above that incentives did not change worker attendance in treatment firms. Surprisingly, attendance among early workers does not improve, given that these workers were in effect only incentivized on attendance, and had the lowest attendance rates in the pre-incentives weeks ([Table A1](#)).¹¹ On the other hand, compliance is driven entirely by female workers, and workers who were consistently late at baseline – female workers are 15 percentage

¹¹37% of worker absences are because of social functions, followed by 19% because there was no work at the firm, 18% because either the worker or a family member was sick, and 10% because of local temple/religious festivals. A majority of these reasons relate to importance of social/kinship relations, as emphasized in [Breza and Kaur \(2025\)](#), but it is worth noting a large fraction of absenteeism attributable to low labor demand. This matches the rate of all-day firm closures recorded in the data - 10% of firms were closed for the entire day. 40% of closures were because the firm did not have work, followed by 36% because of local temple festivals or the firm owner’s personal commitments, and 10% because of power cuts.

points more likely than male workers to comply (column 3); “late” and “very late” workers are 24 and 31 percentage points respectively more likely than “early” workers to comply (column 4). This is expected, since the incentives were more binding on workers who consistently arrived after the preferred start-time, which was more common among women. In fact, this suggests that incentives may reduce the gap in work hours, and potentially earnings between men and women reported earlier (Table A1).

Do workers compensate for arriving earlier at the firm by working fewer hours, or taking more breaks during the work day? Table 5 reports treatment effects on daily work hours, the number and duration of breaks taken in a day. Daily work hours is defined as the total hours worked in a day, excluding the total time spent on breaks. Breaks include any time taken for lunch, tea, or other reasons for which the worker left the firm.¹² All outcomes are coded 0 on days that a worker was absent. Workers in treatment firms work 0.40 additional hours, or 24 additional minutes, relative to workers in control firms (column (1)), who work 6.3 hours per day on average pre-incentives. Similarly, I do not find any statistically or economically significant changes in the number of breaks (column (2)) or total duration of breaks (column (3)).

The treatment effects on work hours and break duration vary slightly across male and female workers, and “early” versus “late” workers; however, most of these differences are small in magnitude and statistically insignificant, with the exception of “late” workers, who increase daily work hours by 0.60 hours, or 36 minutes (statistically significant at the 10% level). This is not surprising when looking at the treatment effect on deviation from the preferred start time, reported in Table A2. On average, workers in treatment firms arrived 12 minutes earlier relative to control workers, but this effect is driven by “late” and “very late” workers (column (3)) – “late” and “very late” workers arrive 11 minutes and 33 minutes earlier relative to control workers.

5.1.1 Does treatment improve worker coordination in arrival times?

Next, I examine how treatment affects worker coordination at the *firm-day level*, conditional on being present. Table A3 reports treatment effects for three measures of dispersion in worker arrival times at the firm-day level: minutes between the 1st and last worker to arrive at the firm (column (1)), the 90-10 gap in worker arrival times (relative to the preferred start time) (column (2)), and the deviation from preferred start time for the 90th percentile worker (columns (3)). This regression excludes firm-days on which only 1 worker was present at the firm, since the notion of coordination is not meaningful in such cases.

Total dispersion in worker arrival times is 133 minutes on average at baseline – treatment reduces dispersion by 15 minutes, a 11% reduction in dispersion, or improvement in coordination relative to control firms (column (1)). The 90-10 gap is a more conservative measure of dispersion which excludes extreme outliers – workers who arrive extremely early or extremely late. Treatment

¹²Workers rarely perform other work for the firm during breaks (5% of worker-days).

further reduces the 90-10 gap by 19 minutes, a 16% decrease relative to the control mean of 119 minutes pre-incentives. Finally, I look at treatment effects on deviation from the owner’s preferred start time for the 90th percentile worker. This measure allows me to test whether these effects are driven by workers for whom the incentive was binding. Indeed, the deviation of the 90th percentile work reduces by 23 minutes in treatment firms, relative to the control group mean of 71 minutes pre-incentives. These effects amount to a 32% reduction in dispersion, or improvement in coordination as a result of treatment.

5.2 Worker Output and Earnings

Next, I test whether incentives increase worker output and earnings. Worker output is measured as the total quantity of pieces stitched by the worker in a week, across all tasks (and products). Note that this does not necessarily correspond to firm output, since a worker who stitched 100 collars and 100 sleeves would be considered to have worked on 200 pieces on that day. However, this measure aligns best with workers earnings, which are constructed as the product of the task specific piece rate and quantity produced on that task, summed over all tasks and products worked on in a week. Output and earnings are coded as 0 on days that a worker was absent, before aggregating to the weekly level. Since firms produce different types of products, I normalize weekly worker output and earnings to pre-incentive, product level, weekly means and standard deviations.

Worker Output Results for weekly worker output are reported in columns (1)-(3) of [Table 7](#). The treatment effect on weekly worker output is 0.35 standard deviations and statistically significant at the 1% level. This amounts to a 35% increase in weekly output relative to the control group mean (column (1) of [Table 7](#)). The difference effects on weekly output among females, or “late” and “very late ” workers compared to their male/early counterparts is not statistically significant (columns (2)-(3) in [Table 7](#)). These results look similar when I use actual levels of output by product type, including product fixed effects to control for level differences by product type (see ??). In levels, workers in treatment firms produced on average 600-1,100 additional pieces per week relative to the average of 3,300-3,700 in control firms, depending on how I aggregate output over different product types.

Worker Earnings Results for weekly worker earnings are reported in columns (4)-(6) of [Table 7](#). The treatment effect on worker earnings mirrors the results on worker output, albeit are smaller in magnitude. This is consistent with the average piece rate per task of INR 0.87. Weekly earnings increase by 0.27 standard deviations (24% of the control mean), significant at the 5% level. Notably, treatment partly closes the weekly earnings gap between early and very late workers in pre-incentive weeks, as documented in [Table A1](#).

Surprisingly, and unlike the results on worker compliance, worker output and earnings are not only concentrated among either female or late and very late workers. This may be due to several reasons, which will determine further analysis and mechanism tests in the paper. Specifically, I

hypothesize two reasons explaining the result on worker earnings: first, by changing worker arrival times (and consequently, both better coordination in arrival times, and a different order in which workers arrive), treatment may have resulted in a change in task allocation between workers in the firm. It is an open question whether the new task allocation is more efficient for workers and firms. I plan to investigate this further by exploring whether new task allocations are better aligned with worker’s expertise¹³, or increase the degree of task specialization in the firm. Second, the fact that workers for whom incentives were not binding also benefit from incentives suggests the presence of coordination externalities along the production. For example, increased compliance among late and/or female workers increases their own output each day, but may have spillover benefits to downstream workers (workers who perform the task next in the order). The firm may realize increases in production through the production line even without changes in the task allocation, or changes in the degree of task specialization. These results are work-in-progress and will be included in future revisions of the paper.

5.3 Total Firm Output and Revenue

How do the changes in individual worker output add up to total firm output and revenues? I construct total firm output using daily production data collected from workers. Total firm output is constructed by adding up total quantity produced on the last task in the production line, across all workers, for each distinct product. Total output is summed by broad product category over the week, and normalized to pre-incentive, product level, weekly means and standard deviations.

I calculate weekly firm revenue as the product of total firm output and the total piece rate that owners earn, and weekly labor cost as the product of total firm output and the total piece rate that workers earn (across the product’s constituent tasks) in each week. Finally, to absorb week by week variation and improve the precision of my estimates, I average total weekly production, revenues and labor costs across the 2 pre-incentive weeks (before) and 4 during-incentive weeks (during), and estimate a double-difference version of [Equation 2](#).

[Table 8](#) reports results on weekly firm revenues (column (1)), total labor costs (columns (2)-(3)), average owner and worker piece rates (columns (4)-(5)) and standardized weekly output (column (6)).

Weekly firm revenues increase by INR 12,000, a 12% increase relative to the control group. Total labor costs including the incentives paid to workers (column (3)) increase by INR 6,200 (15%). There is a small increase in average piece rates that owners receive in treatment firms, but the increase is not statistically significant. Finally, weekly firm output increases by 0.24 SD,

¹³Every week, I record worker productivity at the task they are performing at that time. Productivity is measured as the number of pieces of that task completed within 5-minutes and is recorded by surveyor observation. I also elicited which tasks workers consider themselves to be an expert at in worker baseline surveys. I plan to use these data to assess whether treatment improves task allocation by making workers more likely to work on their expert tasks.

relative to the control mean and standard deviation, this is a 29% increase in total weekly firm output (control mean = 6,900, SD = 7,200). This effect is statistically significant at the 10% level, and is consistent with the magnitude of the effect on individual worker output. These results suggest that improving worker coordination increases firm output and profits, assuming that other fixed, non-labor costs did not change differentially between control and treatment firms. Further analysis will use self-reported profits, revenues, and non-labor costs from the owner endline survey to comprehensively assess the profitability of incentives, and compare these estimates to total weekly payouts made to workers to encourage them to arrive on time.

5.4 Why were owners not incentivizing timely arrival before the experiment?

A natural question arises – if incentives were profitable to firms, why did owners not offer such incentives in the status quo? I provide some suggestive reasons based on survey responses and anecdotes from the field.

First, the existing equilibrium in which these firms operate make it hard for a single firm to enforce common work start times without raising wages or incentivizing workers. Firms are located in dense clusters, where over 95% of firms pay workers on piece rate contracts, and 80% owners report being okay with workers arriving 30 or more minutes late. When asked about the consequences of enforcing common work start times without incentives, owners believe that workers may leave the firm/refused to work at the firm (52%), or they may be less willing to work extra hours when there are urgent orders (38%), and that it may harm their relationship with workers (31%). These reasons point to a combination of hiring frictions, idiosyncratic nature of demand and reputation concerns for owners.

Second, owner's may have incorrect beliefs about workers' response to such incentives, or incorrect beliefs about the profitability of such incentives. In endline surveys, treatment firm owners are more likely to agree that their firm's total production and profits increased during the incentive weeks. This suggests that the experiment itself may have corrected owners' beliefs about the viability of such incentives. 76% of treatment owners reported that they would be willing to continue the incentives if the research team continued to pay for them, indicating that most owners found the incentives were useful. However, only 26% reported that they would be willing to continue the incentives by paying themselves. On average, treatment owners say that they would be willing to contribute INR 30 per day out of INR 100 for such incentives, and 19 out of 40 owners reported 0, indicating a low willingness to pay.

Finally, it may in fact be too costly to incentivize workers in the long-run. While this paper demonstrates the possibility of gains in the short-run, it is not clear whether workers may continue to arrive on time if incentives were to continue indefinitely. The long-run response of workers depends on several other factors. In particular, missing or thin markets for services such as childcare, transportation (public transit, school buses), or the quality of public infrastructure (municipal

water supply, regularity of government ration shops) affect workers' time-use outside of work, and are a key determinant of long-run responses. Perhaps low reported willingness to pay of the owners reflects their beliefs about long-run responses from workers.

Another important long-run consideration is contracting frictions. 7 out of 40 treatment firm owners said that they would be willing to contribute the full amount of the incentive that I offered. However, in conversations with owners after the endline survey, many expressed concern about being able to implement such incentives themselves. Their reasons ranged from contracting frictions to concerns about fairness within the firm. For example, one owner said: *"I'm willing to offer up to Rs. 200, but if workers get used to it, they may start considering it as part of their salary and still not come on time."* Another said *"workers did not feel upset because the incentive came from a third party. If I implement the same system, workers would perceive it as partiality."* The existence of contracting frictions in developing countries is well-known, and is key to understanding why owners were not implementing such incentives before the experiment.

6 Conclusion

This paper experimentally tests whether flexibility in work schedules affects firm productivity in the context of a garment manufacturing cluster in India. I document that workers exercise considerable flexibility in when they arrive at work – at baseline, the 90-10 gap in worker arrival times is almost 2 hours. This stands in contrast to firm owners' preference to have workers arrive at a common time, owing to the production line organization of firms.

I conduct an experiment with 97 small-medium firms where workers in treatment firms are incentivized to arrive before the owner's preferred start time for a period of four weeks. Incentives increase timely arrival by 40%, resulting in a 35% decrease in dispersion in arrival times at the firm level. This is a meaningful effect given the small size of these firms – incentives increase the fraction of workers who arrive within the first 45 minutes of opening time from 60% (3 out of 5 workers) to 80% (4 out of 5 workers).

As a result of the incentives, average weekly revenues increase by 12% in treatment firms, and the increase in revenues exceeds the increase in labor costs *including* incentives. This suggests that offering such incentives could be profitable for firms. However, only 26% owners said they would be willing to offer such incentives themselves. On average, owners are willing to pay INR 30 per day for such incentives, less than a third of the incentives that I offered in the experiment, indicating low willingness to pay.

Survey responses and anecdotal evidence suggests several possible constraints explaining why willingness to pay is low, and why owners were not implementing such incentives in the status quo. These reasons range from incorrect information and beliefs to norms, missing markets and contracting frictions that are prevalent in many low-middle income contexts. The latter three may

require larger scale policy efforts to shift owners' willingness to change the types of contracts and incentives offered to workers. My results demonstrate the importance of policies aimed at improving supply of and access to childcare, and improving the quality of public services such as transportation and municipal residential services (e.g. water supply) as complementary to industrial policies. This also motivates the need for future research to understand what drives the nature of employment contracts offered by firms in developing countries, and their effects on firm productivity.

Finally, it is also striking that incentives did not change worker attendance, and that take-up does not increase to 100% workers arriving on time conditional on attendance. This is despite the incentives being 15% of average daily earnings, indicating that some workers very highly value flexibility. In other survey work with workers across Tirupur district, I document that the provision of flexibility varies systematically with firm size. – small and medium firms are significantly more likely than large firms to offer their workers flexibility in work schedules. Consequently, there is sorting of workers into small (≤ 20 workers) versus large (> 20 workers) firms along gender, marital status and having children – characteristics that positively correlated with demand for flexibility. While workers who work in small firms enjoy flexible work schedules, they do so at the cost of variable pay and lack of access to formal government benefits, such as state provided health insurance, social security, etc. To the extent that demand for flexibility affects workers' labor supply choices, such as whether to be self-employed, work as casual day labor (Oh et al., 2024; Cefala et al., 2024), or work in small firms without formal benefits, this can have important implications for the rate of wage-employment in developing countries. More research is needed to understand drivers of the demand for flexibility in developing countries, as well as broader economic implications.

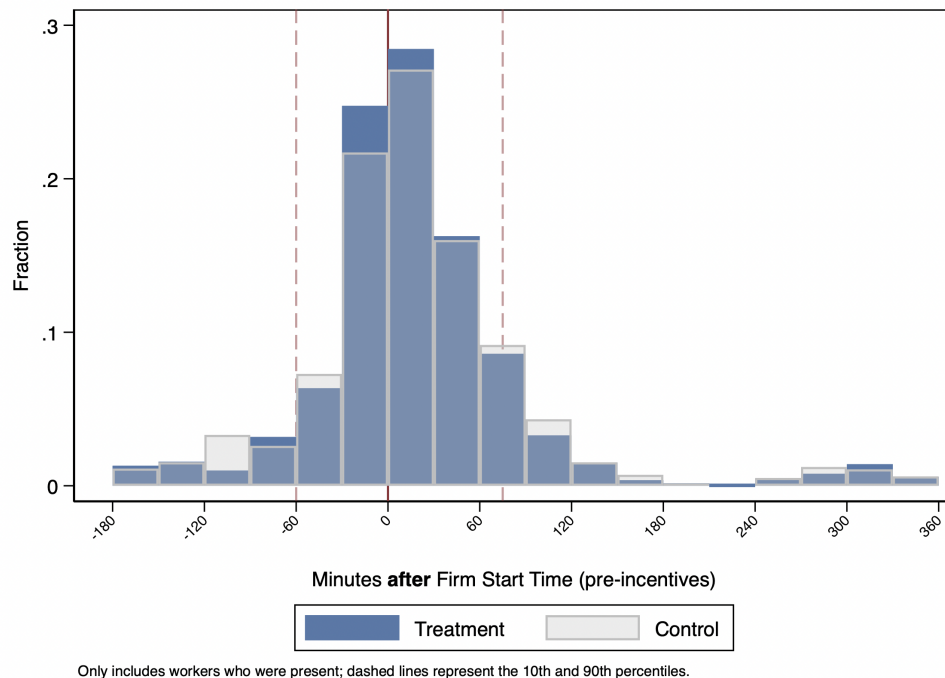
References

- Adhvaryu, A., N. Kala, and A. Nyshadham (2022). Management and shocks to worker productivity. *Journal of Political Economy* 130(1), 1–47.
- Atkin, D., A. Chaudhry, S. Chaudhry, , A. K. Khandelwal, and E. Verhoogen (2017). Organizational barriers to technology adoption: Evidence from soccer-ball producers in pakistan. *The Quarterly Journal of Economics* 1101, 1164.
- Bandiera, O., A. Elsayed, A. Heil, and A. Smurra (2022). Economic development and the organisation of labour. *Working Paper*.
- Banerjee, A. and K. Munshi (2004). How efficiently is capital allocated? evidence from the knitted garment industry in tirupur. *The Review of Economic Studies* 71(1), 19–42.
- Bassi, V., J. H. Lee, A. Peter, T. Porzio, R. Sen, and E. Tugume (2023). Self-employment within the firm. Technical report, National Bureau of Economic Research.
- Bassi, V., R. Muoio, T. Porzio, R. Sen, and E. Tugume (2022). Achieving scale collectively. *Econometrica* 90(6), 2937–2978.
- Bassi, V. and A. Nansamba (2022). Screening and signalling non-cognitive skills: experimental evidence from uganda. *The Economic Journal* 132(642), 471–511.
- Baysan, C., M. H. Dar, K. Emerick, Z. Li, and E. Sadoulet (2024). The agricultural wage gap within rural villages. *Journal of Development Economics* 168, 103270.
- Breza, E. and S. Kaur (2025). Labor markets in developing countries. *Annual Review of Economics* 17.
- Bruhn, M. and D. McKenzie (2009). In pursuit of balance: Randomization in practice in development field experiments. *American economic journal: applied economics* 1(4), 200–232.
- Caria, S. A. and P. Falco (2024). Skeptical employers: Experimental evidence on biased beliefs constraining firm growth. *Review of Economics and Statistics* 106(5), 1352–1368.
- Carranza, E., R. Garlick, K. Orkin, and N. Rankin (2020). Job search and matching with two-sided limited information.
- Cefala, L., S. Kaur, H. Schofield, and Y. Shamdasani (2024). Habit formation in labor supply.
- Cefala, L., P. Naso, M. Ndayikeza, and N. Swanson (2024). Under-training by employers in informal labor markets: Evidence from burundi. Technical report, Working Paper.
- Chaudhry, T. and C. Woodruff (2013). Incentives and productivity: Work groups vs. production lines. *IGC Working Paper (F-6005-PAK-1)*.

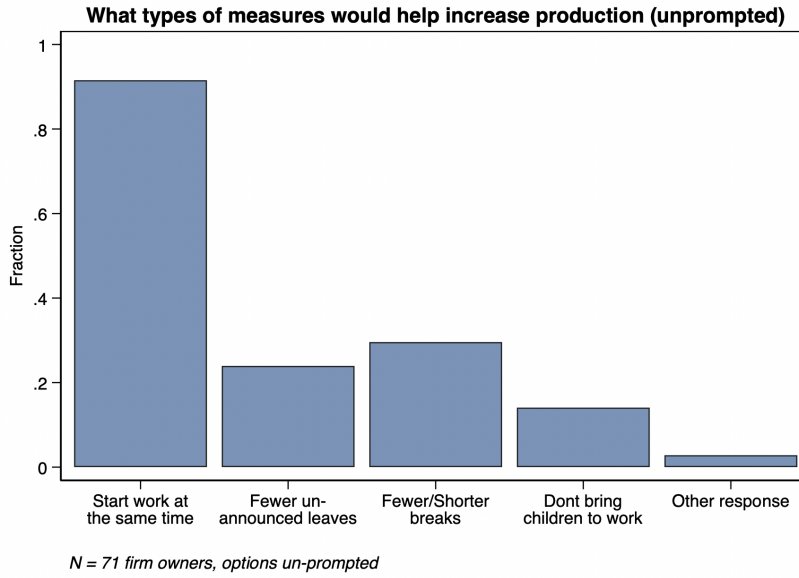
- De Mel, S., D. McKenzie, and C. Woodruff (2019). Labor drops: Experimental evidence on the return to additional labor in microenterprises. *American Economic Journal: Applied Economics* 11(1), 202–235.
- Hardy, M. and J. McCasland (2023). Are small firms labor constrained? experimental evidence from ghana. *American Economic Journal: Applied Economics* 15(2), 253–284.
- Ho, L., S. Jalota, and A. Karandikar (2024). Bringing work home: Flexible work arrangements as gateway jobs for women in west bengal. Technical report, Technical Report.
- Ho, Y. and Y. Huang (2024). Selection and sorting when supervisors have discretion: Experimental evidence from. *Working Paper*.
- Lucas, R. E. (1978). On the size distribution of business firms. *The Bell Journal of Economics*, 508–523.
- Mas, A. and A. Pallais (2017). Valuing alternative work arrangements. *American Economic Review* 107(12), 3722–3759.
- Mas, A. and A. Pallais (2020). Alternative work arrangements. *Annual Review of Economics* 12(1), 631–658.
- Nadvi, K. and H. Schmitz (1994). *Industrial clusters in less developed countries: review of experiences and research agenda*, Volume 39. Institute of Development Studies Brighton.
- Oh, S., Y. Shamdasani, and S. Goraya (2024, September). Worker absences and demand for flexible contracts.
- Wiswall, M. and B. Zafar (2018). Preference for the workplace, investment in human capital, and gender. *The Quarterly Journal of Economics* 133(1), 457–507.

7 Figures

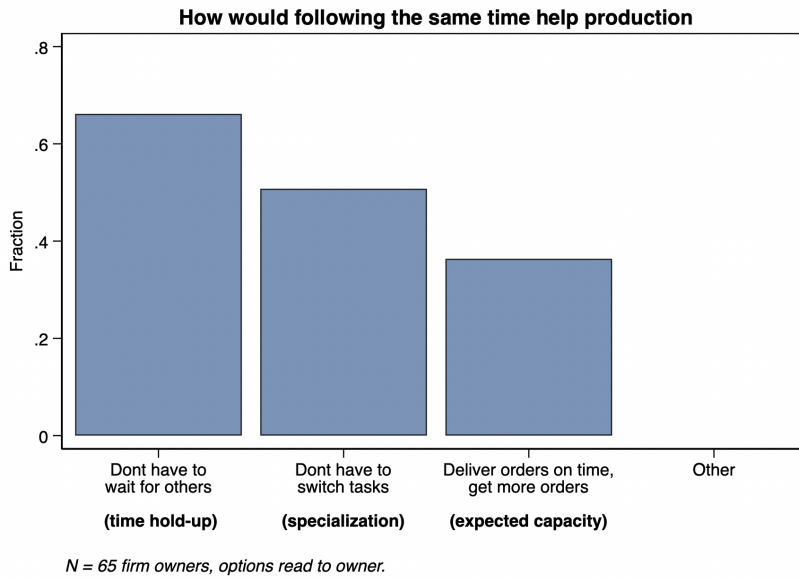
Figure 1: Dispersion in Worker Arrival Times, before Incentives



Notes: This figure plots the distribution of the deviation of worker arrival times from the owner's preferred start time. $N = 97$ firms, ≈ 600 workers.



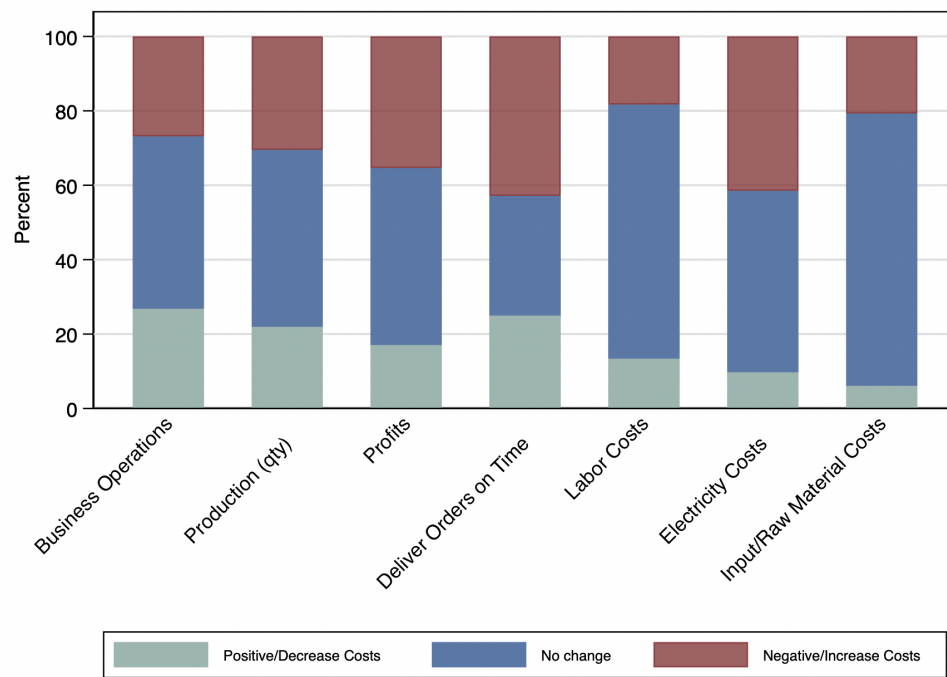
(a) Measures that could help increase production



(b) Why is coordination in worker arrival times important

Notes: Panel A reports owner's responses from the question "What type of measures would help increase production?", with options unprompted. $N = 71$. Panel B reports owner's responses from the question "If everybody starts work at the same time, how will it improve production?", with options 1 = Workers don't have to wait for others, 2 = Workers don't have to switch tasks, 3 = Helps me fulfill orders on time and get more orders, and 4 = Others.

Figure 3: How Providing Flexibility Affects Firm Operations



Notes: This figure reports owner's responses from the question: "Now I will list 4 different business objectives. For each of these, do you think providing workers flexibility affects this objective positively, negatively, or has no effect?" This data is from the full sample of 97 firm owners.

Figure 4: Experiment Design

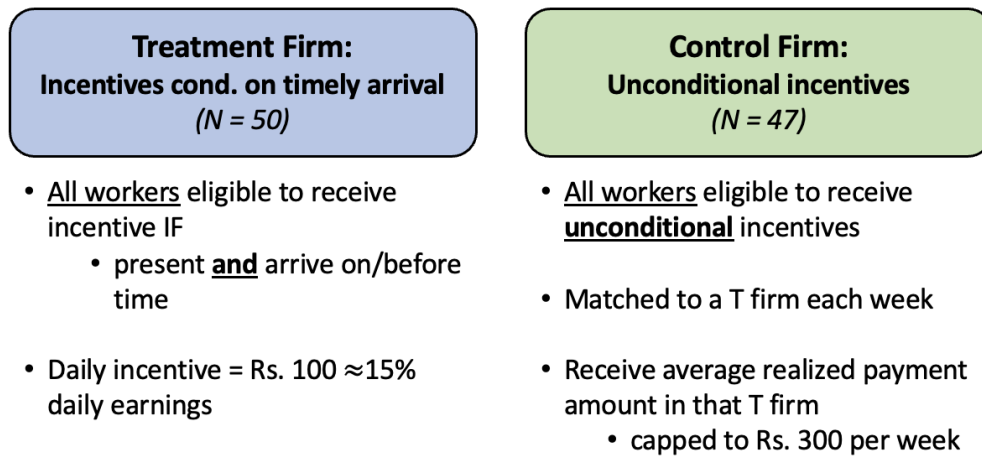


Figure 5: Study Timeline

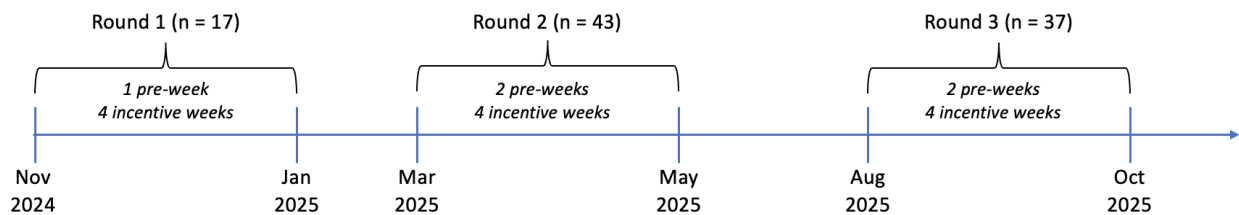


Figure 6: Change in deviation from owner's preferred start time

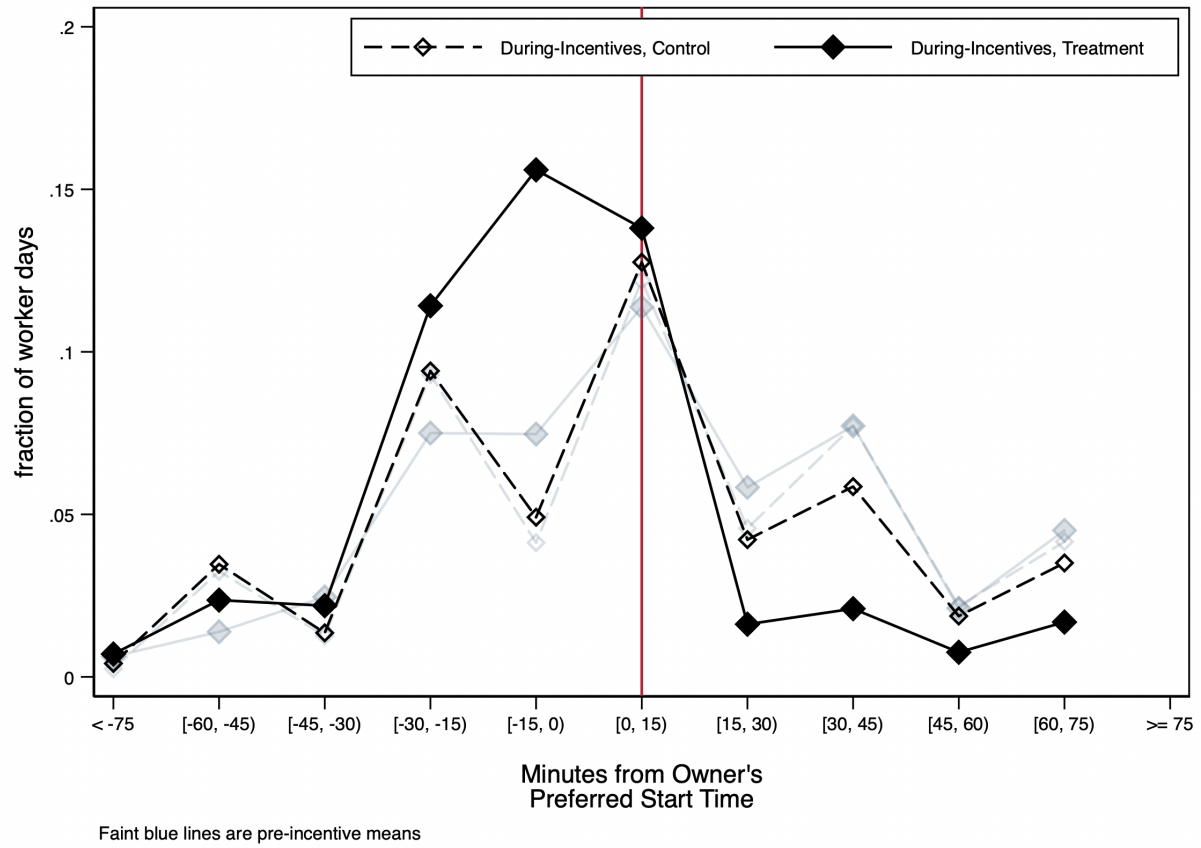
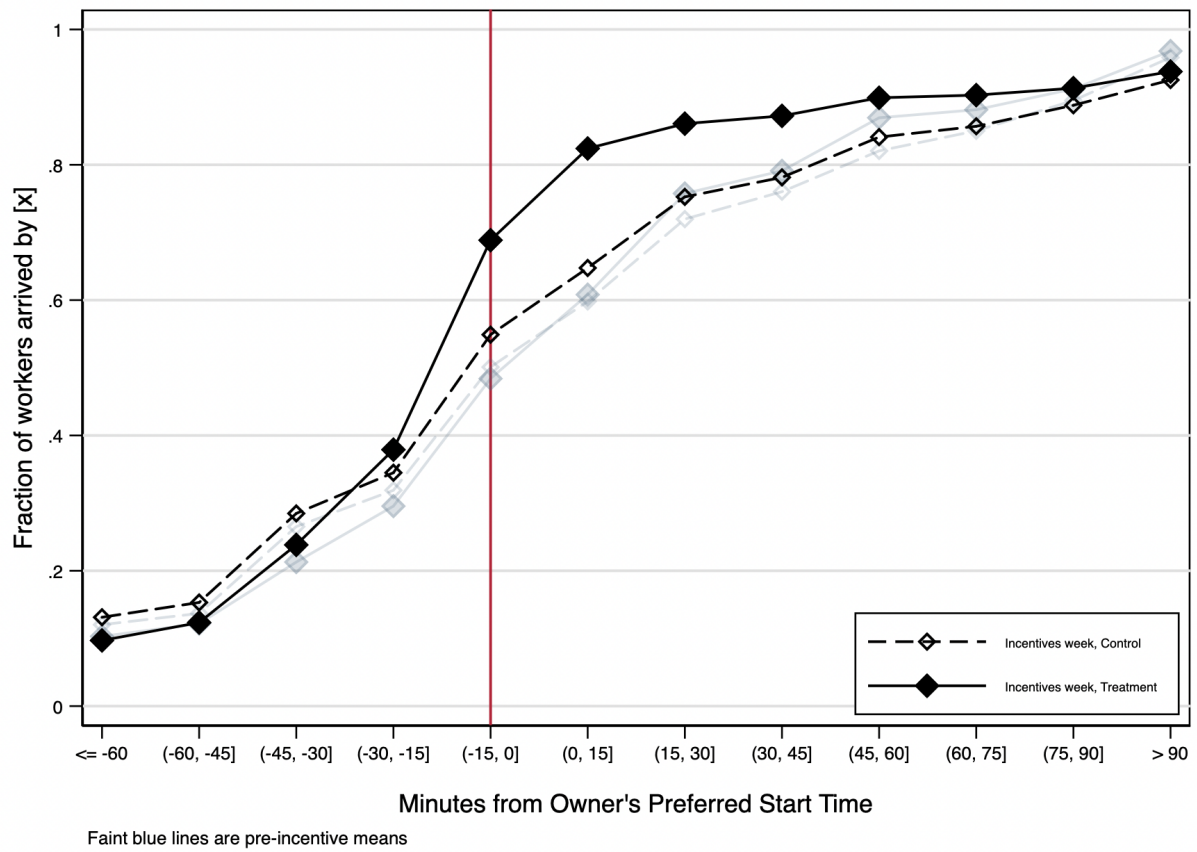


Figure 7: CDF of fraction of workers arriving in 15-minute intervals



8 Tables

Table 1: Balance on Baseline Firm Characteristics

	Control	Treatment	T - C	<i>p-value</i>
	Mean/SD	Mean/SD	Difference	
Firm Characteristics				
# Tailors	6.23 (3.14)	6.06 (3.12)	-0.13	0.84
# Helpers	1.64 (1.15)	1.28 (1.14)	-0.37	0.11
Total # Machines	11.64 (4.86)	11.18 (3.69)	-0.36	0.66
Owner directly supervises workers	0.91 (0.28)	0.98 (0.14)	0.06	0.18
Firm Age (yrs)	6.68 (5.29)	6.55 (6.14)	-0.17	0.89
Export orders	0.13 (0.34)	0.18 (0.39)	0.05	0.47
Weekly Output	4075.61 (2320.43)	4306.98 (2876.29)	286.90	0.59
Revenues (last 30 days)	134444.19 (118877.68)	134514.29 (113349.76)	-257.95	0.99
Profits (last 30 days)	33962.98 (58396.33)	27632.65 (21124.33)	-6454.14	0.47
Total Costs - wage bill (last 30 days)	32281.66 (20389.70)	33018.80 (18815.08)	973.06	0.81
Owner Demographics				
Age	39.21 (8.36)	37.60 (8.79)	-1.57	0.35
Male	0.85 (0.36)	0.82 (0.39)	-0.03	0.73
Completed Secondary School	0.57 (0.50)	0.56 (0.50)	-0.01	0.91
Commute Time (mins)	6.32 (6.66)	8.56 (8.98)	2.26	0.16
Experience in Industry (yrs)	19.15 (10.47)	17.96 (9.65)	-1.36	0.50
Owner does stitching work	0.81 (0.40)	0.78 (0.42)	-0.03	0.74
Owner stitches as a regular tailor	0.61 (0.49)	0.54 (0.50)	-0.06	0.51
Observations	47	50		

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with robust standard errors.

Table 2: Balance on Baseline Worker Characteristics

	Control	Treatment	T - C	<i>p-value</i>
	Mean/SD	Mean/SD	Difference	
Age	37.22 (8.75)	36.35 (9.05)	-0.95	0.31
Male	0.45 (0.50)	0.44 (0.50)	-0.01	0.89
Married	0.94 (0.43)	0.91 (0.46)	-0.04	0.35
Has children < 14 yrs	0.89 (1.26)	0.85 (1.15)	-0.05	0.72
Has elderly member in HH	0.36 (0.48)	0.39 (0.49)	0.03	0.49
Related to owner	0.21 (0.41)	0.21 (0.41)	0.00	0.99
Native to TN	0.92 (0.27)	0.94 (0.24)	0.02	0.51
Commute < 15 mins	0.87 (0.34)	0.84 (0.37)	-0.03	0.39
Completed Secondary School	0.45 (0.50)	0.40 (0.49)	-0.05	0.28
Experience in Industry (yrs)	15.52 (9.25)	14.16 (9.53)	-1.45	0.12
Tenure at Firm (yrs)	2.57 (3.60)	2.39 (3.18)	-0.16	0.73
Observations	304	306		

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with standard errors clustered at the firm level.

Table 3: Worker Attendance and Compliance

	Present	Present and Arrived on Time	Arrived on time, if Present
	(1)	(2)	(3)
Treatment $\times$ During Incentives	0.03 [0.03]	0.18*** [0.03]	0.21*** [0.03]
Control Mean (Pre-Incentives)	.71	.41	.57
Control Mean (Incentive Weeks)	.75	.48	.65
Treatment Effect %	4.37	42.81	36.22
Observations	16881	16881	12597

Notes: All specifications include worker, day-of-week, week and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level. Attendance is defined as an indicator for whether a worker was present at the firm on that day. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Heterogeneity in Treatment Effects on Compliance and Attendance

	Attendance		Compliance		Work Hours	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	0.02 [0.03]	0.03 [0.04]	0.09** [0.04]	0.01 [0.04]	0.13 [0.32]	0.04 [0.37]
Treatment $\times$ During $\times$ Female	0.01 [0.04]		0.15*** [0.05]		0.42 [0.37]	
Treatment $\times$ During $\times$ Late		0.03 [0.04]		0.24*** [0.05]		0.56 [0.42]
Treatment $\times$ During $\times$ Very Late		-0.02 [0.05]		0.31*** [0.06]		0.66 [0.56]
Control Mean (Pre-Incentives)	.71	.71	.41	.41	6.32	6.32
Omitted Group Mean (Pre-Incentives)	.73	.62	.56	.55	6.78	5.81
Treatment Effect %	4.75	7.62	58.1	60.8	8.68	9.39
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0	0.36		0.00		0.17	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0		0.12		0.00		0.09
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0		0.80		0.00		0.20
Worker FE						
Observations	16571	16881	16571	16881	16295	16597

Notes: All specifications include worker, day-of-week, week, and phase $\times$ strata fixed effects, from the specification in Equation 1. Standard errors are clustered at the firm level. Attendance is defined as an indicator for whether a worker was present at the firm on that day. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Treatment Effects on Daily Work Hours

	Work Hours	Total # Breaks	Total Break Duration (mins)
	(1)	(2)	(3)
Treatment $\times$ During Incentives	0.40 [0.31]	0.08 [0.10]	0.85 [3.47]
Control Mean Pre-Incentives	6.32	2.24	57.65
Control Mean Incentive Weeks	6.88	2.08	60.93
Treatment Effect %	6.29	3.62	1.47
Observations	16597	14395	16881

Notes: All specifications include dow, week and phase x strata fixed effects. Standard errors are clustered at the firm level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Heterogeneity in Treatment Effects on Daily Work Hours

	Work Hours		Total # Breaks		Total Break Duration (mins)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	0.13 [0.32]	0.04 [0.37]	0.11 [0.10]	0.06 [0.11]	0.37 [4.38]	-0.37 [4.34]
Treatment $\times$ During $\times$ Female	0.42 [0.37]		-0.06 [0.10]		0.06 [4.02]	
Treatment $\times$ During $\times$ Late		0.56 [0.42]		0.07 [0.13]		4.62 [4.72]
Treatment $\times$ During $\times$ Very Late		0.66 [0.56]		0.01 [0.17]		-0.33 [4.34]
Control Mean (Pre-Incentives)	6.32	6.32	2.24	2.24	57.65	57.65
Omitted Group Mean (Pre-Incentives)	6.78	5.81	2.3	2.28	60.63	50.71
Treatment Effect %	8.68	9.39	2.46	5.5	.75	7.37
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0	0.17		0.67		0.91	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0		0.09		0.31		0.31
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0		0.20		0.68		0.88
Worker FE	X	X	X	X	X	X
Observations	16295	16597	14303	14395	16571	16881

Notes: All specifications include day-of-week, week, phase $\times$ strata and worker fixed effects, from the specification in Equation 1. Standard errors at clustered at the firm level. Work hours are calculated as the difference between worker's exit and entry times, subtracting the total time spent on lunch, tea and other breaks during which the worker was not stitching. Total # of breaks includes the number of lunch, tea and other breaks taken by the worker during the day, and total break duration (in minutes) is the sum of time spent across all work breaks in a day.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Normalized Worker Output and Earnings

	Weekly Output (Qty, normalized)			Weekly Earnings (INR, normalized)		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	0.35*** [0.13]	0.31** [0.13]	0.21 [0.15]	0.27** [0.12]	0.26* [0.15]	0.22 [0.15]
Treatment $\times$ During $\times$ Female		0.08 [0.10]			0.03 [0.12]	
Treatment $\times$ During $\times$ Late			0.27 [0.19]			0.09 [0.19]
Treatment $\times$ During $\times$ Very Late			0.19 [0.21]			0.03 [0.23]
Control Mean (Pre-Incentives)	3455.83	3455.83	3455.83	2861.61	2861.61	2861.61
Control SD (Pre-Incentives)	3456.07	3456.07	3456.07	2610.07	2610.07	2610.07
Omitted Group Mean (Pre-Incentives)		3597.22	3228.91		3292.65	2980.11
Treatment Effect %	35.31	39.04	48.19	24.44	26.59	27.78
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.01			0.02	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.01			0.07
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.04			0.25
Worker FE	X	X	X	X	X	X
Observations	1831	1768	1829	1831	1768	1829

Notes: SEs clustered at firm level. All specifications include day-of-week, week and phase $\times$ strata FE. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. Output is normalized by product relative to pre-incentive week means. Workers who are absent on a given day are assumed to have worked on the product that most other workers in the firm worked on. Earnings are coded as 0 for workers who are absent on a given day. Output and earnings are winsorized at the 99th percentile before normalizing.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Treatment Effects on Weekly Firm Output

	Revenues (INR)	Total Labor Cost (INR)		Piece Rates (INR)		Pieces (SD)
		w/o subsidy	w/ subsidy	Owner	Worker	
Treatment $\times$ During	12523.46** [5258.03]	5747.66* [3023.62]	6240.98** [3031.01]	1.42 [0.96]	0.13 [0.25]	0.24* [0.14]
Control Mean (Pre-Incentives)	97098.14	40206.33	40206.33	15.7	6.88	6905.23
Treatment Effect %	12.9	14.3	15.52	9.04	1.85	29.26
p-value (Treatment $\times$ During)	0.02	0.06	0.04	0.15	0.61	0.10
Firm FE	X	X	X	X	X	X
Observations	114	114	114	114	114	114

Notes: Outcomes collapsed to weekly average before and during incentives (two t per firm) to reduce noise. SEs in brackets, clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A Appendix Tables

Table A1: Worker Labor Supply and Production Outcomes - Pre Incentives

	All workers (1)	by Avg. Dev. in Arrival Time before incentives		
		≤ 0 (Early) (2)	$\in (0,30]$ mins (Late) (3)	> 30 mins (Very Late) (4)
Fraction Early/Late		0.44 (0.00)	0.26 (0.00)	0.30 (0.00)
Attendance, Arrival Times				
Present	0.72 (0.45)	0.65 (0.48)	0.80 (0.40)	0.76 (0.43)
Complied w preferred start time	0.41 (0.49)	0.58 (0.49)	0.44 (0.50)	0.13 (0.34)
Dev. in Arrival Time (mins)	11.56 (74.68)	-31.92 (53.65)	14.03 (44.73)	67.85 (81.80)
Dev. in Exit Time (mins)	-41.79 (107.64)	-31.91 (105.58)	-47.41 (106.67)	-50.02 (110.25)
Daily Work Hours	6.46 (4.37)	6.24 (4.81)	7.21 (3.93)	6.16 (3.89)
Total # Breaks	2.25 (1.41)	2.25 (1.51)	2.33 (1.22)	2.20 (1.42)
Total Break Duration - mins	59.02 (45.20)	53.52 (46.95)	66.56 (41.74)	61.28 (44.13)
Production				
Avg Task Piece Rate (INR)	0.88 (1.39)	1.04 (1.74)	0.75 (1.03)	0.73 (0.91)
Output per hour	92.67 (99.23)	94.06 (125.34)	97.39 (85.04)	86.54 (64.81)
Daily Earnings (INR)	482.66 (619.50)	525.15 (743.34)	491.79 (530.02)	407.52 (443.11)
(max) task_first_alt	0.16 (0.37)	0.17 (0.38)	0.16 (0.37)	0.15 (0.35)
(max) task_last_alt	0.34 (0.47)	0.36 (0.48)	0.34 (0.47)	0.31 (0.46)
Singer Task	0.19 (0.39)	0.23 (0.42)	0.14 (0.35)	0.17 (0.37)
Complex Singer Task	0.11 (0.31)	0.16 (0.36)	0.07 (0.25)	0.06 (0.23)
Demographics				
Female	0.54 (0.50)	0.32 (0.47)	0.60 (0.49)	0.82 (0.38)
Married	0.82 (0.38)	0.74 (0.44)	0.86 (0.34)	0.93 (0.26)
# children < 14 yrs	0.91 (1.30)	0.80 (0.88)	1.00 (1.57)	1.01 (1.53)
Commute < 15 mins	0.86 (0.35)	0.85 (0.36)	0.84 (0.37)	0.88 (0.32)
Experience in Industry (yrs)	15.19 (9.46)	16.66 (9.94)	14.62 (9.14)	13.49 (8.63)
Owner Baseline Ranking				
Punctual	0.19 (0.39)	0.24 (0.42)	0.18 (0.38)	0.12 (0.33)
Reliable	0.20 (0.40)	0.22 (0.42)	0.25 (0.43)	0.14 (0.34)
Productive	0.19 (0.39)	0.22 (0.42)	0.25 (0.43)	0.10 (0.29)
Knowledgable about Prod.	0.20 (0.40)	0.23 (0.42)	0.21 (0.41)	0.15 (0.36)
would benefit most from intervention	0.21 (0.41)	0.21 (0.41)	0.20 (0.40)	0.23 (0.42)
Observations	5358	2467	1334	1557

Notes: Data at the worker-day level, includes 1-2 weeks of pre-incentives data, excludes days firm was closed. All outcomes coded as 0 if worker was absent. Piece rate and earnings are in INR.

Table A2: Treatment Effects on Deviation from Start Time

	Deviation from Start Time (mins)		
	(1)	(2)	(3)
Treatment $\times$ During	-12.05*** [3.86]	-8.32* [4.32]	3.26 [3.80]
Treatment $\times$ During $\times$ Female		-6.65 [4.38]	
Treatment $\times$ During $\times$ Late			-14.05*** [3.59]
Treatment $\times$ During $\times$ Very Late			-36.91*** [5.23]
Control Mean (Pre-Incentives)	10.66	10.66	10.66
Omitted Group Mean (Pre-Incentives)		-17.41	-34.43
Treatment Effect %	-113	-140.46	-101.19
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.00	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.01
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.00
Worker FE			
Observations	12617	12515	12617

Notes: All specifications include day-of-week, week, phase $\times$ strata and worker fixed effects, from the specification in Equation 1. Standard errors are clustered at the firm level. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Treatment Effects on Coordination - Firm Level

	Mins between 1st and last worker		Mins between the 10th and 90th pctl worker		Deviation from start-time: 90th pctl worker	
	(1)	(2)	(3)	(4)	(5)	(6)
T × During	-15.00 [9.90]	-14.96 [9.90]	-19.34** [8.37]	-19.08** [8.38]	-22.80*** [8.23]	-22.44*** [8.26]
Control Mean Pre-Incentives	132.83	132.83	119.07	119.07	71.01	71.01
Control Mean Incentive Weeks	116.69	116.69	103.38	103.38	53.52	53.52
Treatment Effect %	-11.29	-11.26	-16.24	-16.02	-32.1	-31.6
Observations	2657	2934	2657	2934	2657	2934

Notes: This table reports firm-day level outcomes on coordination in worker arrival times. All specifications include dow, week and phase x strata fixed effects. Standard errors are clustered at the firm level. All outcome variables are in minutes, and are only computed for firm-days where strictly greater than 1 worker was present.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Flexibility by Firm Size

	Small Firms		Large Firms	
	Mean	SD	Mean	SD
Timings Same	0.49	(0.50)	0.77	(0.42)
Timings Strict	0.15	(0.36)	0.50	(0.51)
Entry time is recorded	0.03	(0.17)	0.50	(0.51)
Pay cut if late	0.06	(0.23)	0.40	(0.49)
Margin of Lateness				
Acceptable to be 5 minutes late	0.97	(0.17)	0.94	(0.24)
Acceptable to be ≤ 30 minutes late	0.91	(0.29)	0.69	(0.47)
Acceptable to be > 30 minutes late	0.85	(0.36)	0.62	(0.49)
Observations	70		48	

Notes: SD in parenthesis (even columns). Small firms are defined as firms with up to 20 workers, and where the owner is often both a tailor (either full or part-time) and manager of the firm. Large firms include firms with at least 50 workers, where ownership and factory-floor management are separate.